Software Project Management
Assignment-1
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the following can be considered a job (routine operation) rather than a project?

a. Processing monthly employee salaries
b. Developing a new mobile application
c. Manufacturing cars on an assembly line
d. Organizing the Olympic Games
e. Handling daily customer support requests

Correct Answer: a. Processing monthly employee salaries
c. Manufacturing cars on an assembly line
e. Handling daily customer support requests

Detailed Solution:

Processing monthly salaries, manufacturing cars on an assembly line, and handling daily customer support requests are all repetitive, ongoing operations with well-defined procedures. They lack the temporary, unique, and goal-oriented nature of projects, which have a clear start and end, such as developing a new app or organizing the Olympics.

QUESTION 2:

Software projects are generally considered harder to manage than manufacturing projects. Which of the following are possible reasons for this?

a. Software products are intangible
b. Software development involves highly skilled intellectual work
c. Software projects are always larger than manufacturing projects
d. Requirement changes can have significant effects on software projects
e. Software projects require people management

Correct Answer: a. Software products are intangible
b. Software development involves highly skilled intellectual work
d. Requirement changes can have significant effects on software projects

Detailed Solution:

Software products are intangible, making progress difficult to measure visually, and development relies on highly skilled intellectual work, which is less predictable than manufacturing tasks. Additionally, requirement changes can have a significant ripple effect on software projects due to their malleable nature, whereas manufacturing changes are often costlier to implement later. People management, while important, is required in all projects and is not a unique reason for software's added difficulty.

QUESTION 3:

Which of the following statements about software complexity are true?

a. Professional developers usually make any software that they develop as complex as possible
b. Making changes to software for maintenance purposes does not affect its complexity.
c. Increased software complexity generally makes maintenance more difficult.
d. Repeated modifications can increase software complexity.
e. Software complexity automatically decreases as software ages.
f. Higher software complexity can increase the likelihood of introducing defects during maintenance.

Correct Answer: c. Increased software complexity generally makes maintenance more difficult.
d. Repeated modifications can increase software complexity.
f. Higher software complexity can increase the likelihood of introducing defects during maintenance.

Detailed Solution:

Increased software complexity makes the code harder to understand and modify, directly increasing maintenance difficulty. Repeated modifications over time tend to add structural clutter (often called software entropy), which raises complexity, and this higher complexity increases the chance of introducing new defects when changes are made. Professional developers aim for simplicity, maintenance changes do affect complexity, and complexity does not automatically decrease with age—it typically grows unless actively refactored.

QUESTION 4:

Which of the following statements regarding project stakeholders are true?

a. Sponsors are not considered stakeholders because they only provide resources for the project.
b. A stakeholder can be an individual or an organization.
c. Stakeholders may influence project execution and outcomes.
d. Customers are stakeholders because they use the project's product or service.
e. Only project team members are the stakeholders of the project

Correct Answer: b. A stakeholder can be an individual or an organization.
c. Stakeholders may influence project execution and outcomes.
d. Customers are stakeholders because they use the project's product or service

Detailed Solution:

Stakeholders include any individual or organization that is actively involved in or affected by the project, which covers sponsors, customers, users, and the project team. They can influence project execution and outcomes, and customers are indeed stakeholders because they use or are impacted by the final product. Sponsors are also key stakeholders despite providing resources, and limiting stakeholders to just the project team is incorrect.

QUESTION 5:

Which of the following factors can negatively affect the success of a software project?

a. Frequent changes in business needs.
b. Inexperienced project team members.
c. Effective project management.
d. Rapid changes in the chosen technology.
e. Poor understanding of customer expectations.

Correct Answer: a. Frequent changes in business needs.
b. Inexperienced project team members.
d. Rapid changes in the chosen technology.
e. Poor understanding of customer expectations.

Detailed Solution:

Frequent changes in business needs, inexperienced team members, rapid technology shifts, and poor understanding of customer expectations all introduce uncertainty, rework, and misalignment, which directly harm project success. In contrast, effective project management is a positive factor that helps mitigate risks and keep the project on track.

QUESTION 6:

Which of the following statements is correct regarding various types of software projects?

a. Packaged software is rarely available for the vertical market
b. Packaged software is intended for multiple customers.
c. Custom software is always developed from scratch.
d. Custom software is developed to satisfy specific user requirements.
e. Packaged software is always cheaper than custom software.

Correct Answer: b. Packaged software is intended for multiple customers.
d. Custom software is developed to satisfy specific user requirements.

Detailed Solution:

Packaged software is built for multiple customers with common needs, while custom software is tailored to satisfy specific user requirements for a particular organization. Custom software is not always developed from scratch (it can reuse components), and packaged software is not always cheaper—total costs depend on licensing, customization, and fit. Vertical-market packaged software is also widely available.

QUESTION 7:

Which of the following is not considered to be Embedded Software?

a. TV Remote Software
b. Nuclear Plant Control Software
c. Patient Management Software for a hospital
d. Robot Control Software
e. Automobile Control Software

Correct Answer: c. Patient Management Software for a hospital

Detailed Solution:

Patient Management Software for a hospital runs on general-purpose computers or servers and is used for administrative and clinical data management, making it an information system rather than embedded software. In contrast, TV remote software, nuclear plant control, robot control, and automobile control software all run on specialized hardware (microcontrollers or processors) within larger devices to perform specific control functions, which is the defining characteristic of embedded software.

QUESTION 8:

Which of the following statements regarding project management activities are correct?

a. Planning is performed during execution.
b. Feasibility analysis generally precedes project planning.
c. Project execution involves carrying out planned activities.
d. Feasibility study and project execution are identical activities.
e. A project manager may be involved in all three phases.

Correct Answer: b. Feasibility analysis generally precedes project planning.
c. Project execution involves carrying out planned activities.
e. A project manager may be involved in all three phases.

Detailed Solution:

Feasibility analysis is conducted before detailed project planning to assess viability, and project execution is indeed the phase where planned activities are carried out. A project manager is typically involved across feasibility, planning, and execution phases. Planning is primarily done before execution (though some iterative planning may occur), and feasibility study and execution are completely distinct activities.

QUESTION 9:

Which of the following activities are typically associated with project planning?

a. Formally accepting project deliverables.
b. Developing schedules for project activities.
c. Conducting lessons-learned reviews.
d. Allocating project resources.
e. Releasing project resources after completion.

Correct Answer: b. Developing schedules for project activities
d. Allocating project resources

Detailed Solution:

Developing schedules for project activities and allocating project resources are core project planning activities. Formally accepting deliverables, conducting lessons-learned reviews, and releasing resources all occur during project closing or execution phases, not during planning.

QUESTION 10:

Which of the following are benefits of defining milestones in a software project?

a. Automatic generation of source code.
b. Elimination of the need for project planning.
c. Effective addressing of manpower turnover
d. Improved progress tracking.
e. Better visibility into project status.

Correct Answer: d. Improved progress tracking.
e. Better visibility into project status.

Detailed Solution:

Defining milestones improves progress tracking by providing clear checkpoints to measure achievement against the schedule, and it gives better visibility into project status for stakeholders. Milestones do not generate code, eliminate the need for planning, or directly address manpower turnover—they are management tools for monitoring and communication.

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